ITSM Service Desk Analytics

Project Documentation

Project Overview

An interactive IT Service Management analytics project built with Databricks SQL and Power BI. It explores service desk ticket volume, SLA performance, resolution times, support teams, and customer satisfaction.

The analysis uses a synthetic dataset containing 8,000 support tickets and five related reference tables. No real customer or company records are used.

Tools & Technologies

Databricks SQL • Power BI • Power Query • DAX • CSV files

Data Preparation

Six CSV tables (tickets, users, departments, services, technicians, and assets) were loaded into Databricks. The data was reviewed and connected in Power BI through a relational model. Missing resolution and closure values were retained because they represent tickets that had not yet reached those stages.

Dashboard Pages

Page	Purpose
Executive Service Overview	Main KPIs, monthly ticket trend, priority distribution, SLA status, and average resolution time.
Ticket Drivers & Segments	Tickets by support team, department, category, channel, service group, and impact/urgency.
Insights & Recommendations	First contact resolution, customer satisfaction, reopened tickets, and SLA breach rates.

Key Findings

Metric	Result
Total tickets	8,000
SLA compliance (evaluated tickets)	86.16%
Average resolution time	11.86 hours
Reopened ticket rate	17.35%
P1 Critical SLA breach rate	36.09%

Skills Practiced

Data preparation and validation; SQL querying and exploratory analysis; data modeling; DAX measures and KPI calculations; and Power BI dashboard design and visualization.

Author

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